

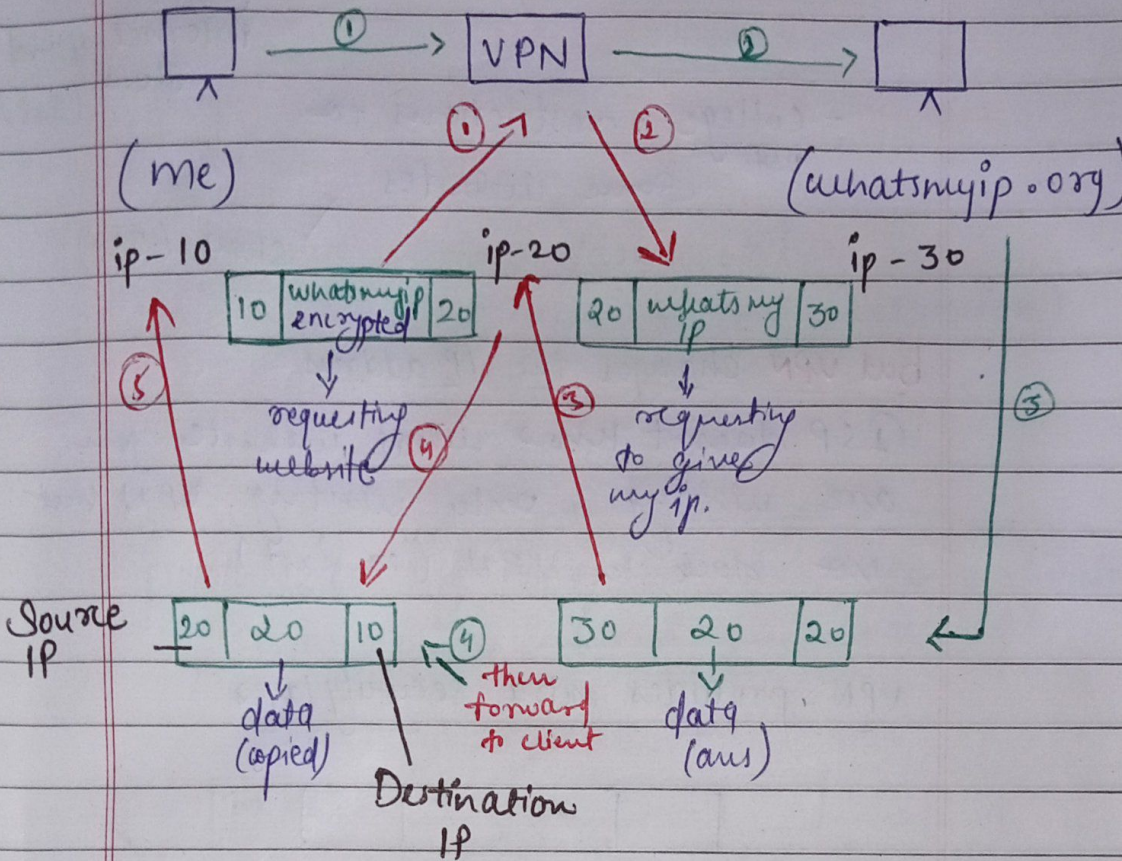
Basic of Internet

Lec 6

Date 23/9/24
Page

Dark Web & Internet Working

→ VPN - IP address in whatsmyip.org



→ **SIP** + **Data encrypted** + **DIP (YT)**

ISP knows SIP & DIP

it knows we visit youtube but not the content.

→ **VPN fast or slow? (- internet)**

usually slow because it is mediator
client → server (faster)

client → VPN → server (comparitively slower)

→ exception (special scenario)

I SP - Streaming videos (access)

Internet speed
Slow (ISP)

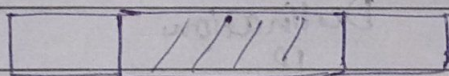
- college : restrictions to
wifi Some websites

slow (ISP)

but VPN changes the IP address

ISP doesn't know what website you
are visiting, only visiting VPN but
no block in VPN (so fast)

VPN provides extra security layer



data

(layer by VPN)

ISP doesn't know what is data

↳ VPN makes it fast

- ISP can't read but it guesses by seeing
data packet (pattern)

VPN can read encrypted request.

↳ Not Safe !!

↳ it can disclose your
identity & search history

So safe VPN must not save logs.
but govt. rule maintain 5 yrs logs

→ Dark web

- illegal
- weapon, murder

↓
Tor

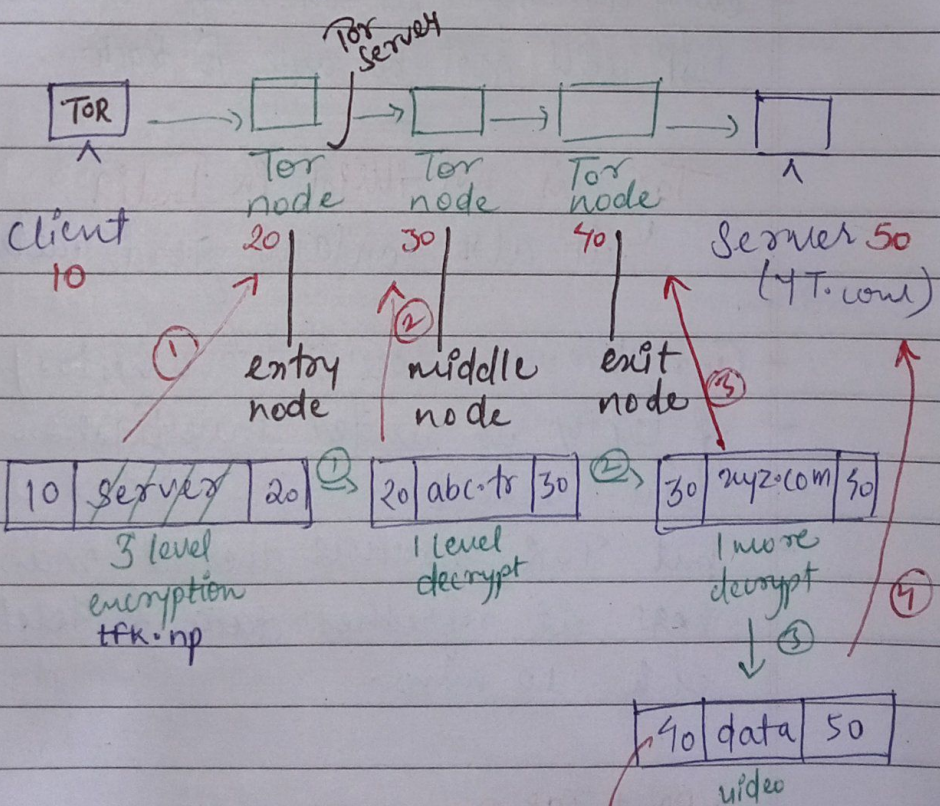
- Scam

browser (access dark web)

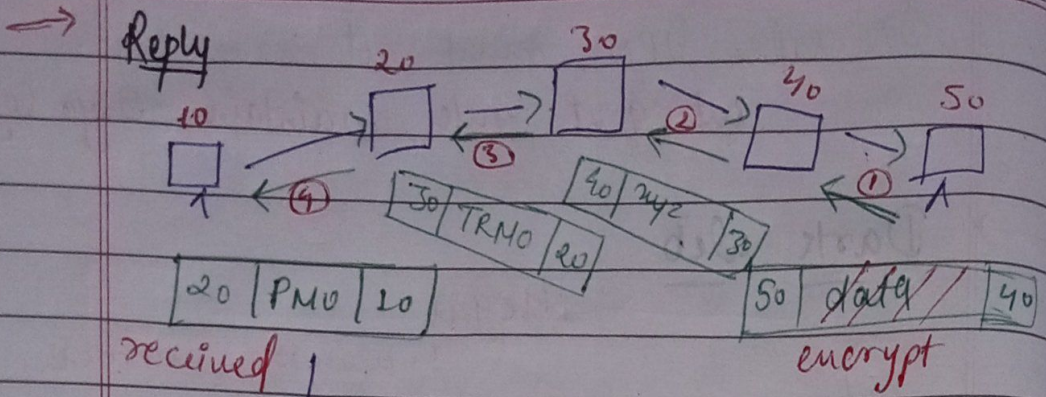
websites → xyz.onion (extension)

anonymous & firstly used by
secure US military

protocols by Tor Browser to maintain
anonymity (not fast internet)



SIP - anonymous



user have authority to decrypt & view data.

ISP → knows you are 10 & you accessed 20 i.e., tor browser, max it can get PMO but it 3 level encrypted.

- every tor node is in diff. countries so ISP will not be able to track.

Tor is not illegal in India

↳ it also contains good websites

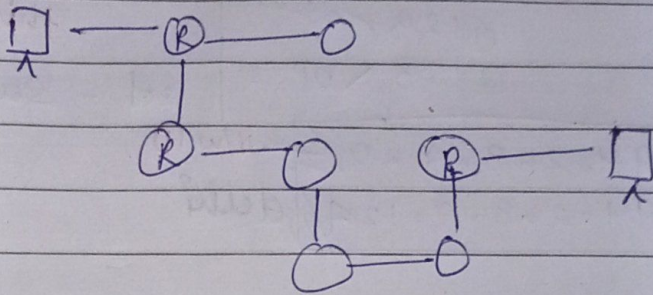
- Tor browser doesn't track history
- if laptop is under investigation then OS can reveal
- but TOR browser doesn't maintain logs it refreshes data & deletes in each 10 mins.

VPN + TOR Browser

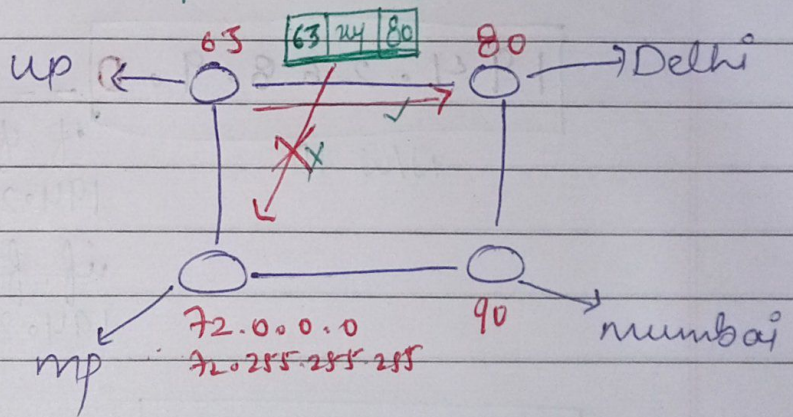
↳ double security

- normally tor uses 3 nodes architecture but for extra security in some case it can be 6 nodes also. (very slow)

Routers decide path by tracking IP address is very slow if IP addresses were random

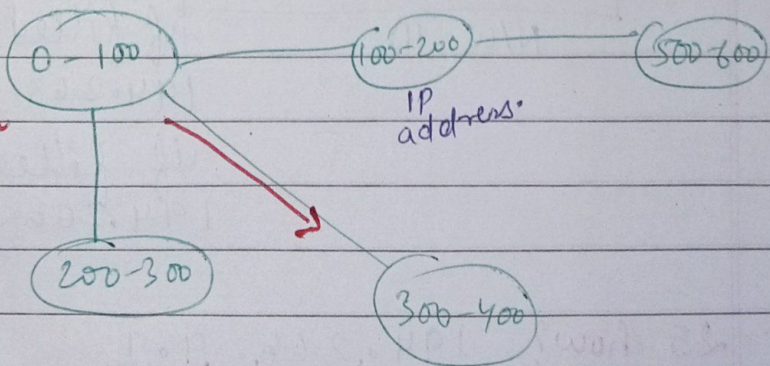


Solution is dividing IP address with same N/w ID for one region.



[340]

early delines



Class A

Subnet Masking

Idea: 73.0.0.0

194.268.9.0

$2^{24-2} = 1.67$ cr.
ip add.

Class C

$2^8 - 2 = 254 / 256 - 2$

Take 256 for now

It can't give it to delhi becoz maybe there are not this much people there.

- ① 128 - delhi
- ② 128 - UP

sol - Subnet masking

① 194.268.9.0 / 25 N/W ID
194.268.9.128 / 25 delhi

N/W ID 194.268.9.128 / 25 UP

② 194.268.9.255 / 25

subnet mask
here 25 bit are set change next 7 bit to get n/w ID

194.268.9.0
N/W ID

if filled 0
194.268.9.0

if filled 1
194.268.9.127

194.268.9.1
N/W ID

if filled 0
194.268.9.128

if filled 1
194.268.9.255

25 how? 194.268.9.1

8bit 8bit 8bit 1bit = 25 bit
 $32 - 25 = 7$ bit

- Router store subnet mask info.

220 . 68 . 90 . 0
└──────────┘ H/ID
24 bit

(IP Address)

0 ← all 0 0 1 all 0 └── 128
to get
n/w ID

220 . 68 . 90 . 0

220 . 68 . 90 . 128